

Americas Energy & Transition

Americas Uranium & Nuclear: Which of the 79 SMR (Small Modular Reactor) projects will win?



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We are uranium & nuclear bulls (e.g., [Bernstein 2026 Uranium/Nuclear Outlook: Glow up or blow up?](#)) and one catalyst around uranium demand we hear is the success of SMRs (Small Modular Reactors). In this note we review and rank the 79 SMR projects in the pipeline, highlight potential commercial winners, but ultimately see other catalysts as more important to uranium demand over the next decade. GEV (Outperform) and CCJ (Outperform) are on the leaderboard for commercial SMR.

We don't think so. We lay our reasons below -

- If the load is large, urgent, and 24/7, the first nuclear answer is likely the existing fleet: life extensions, uprates, restarts, and eventually large reactors. SMR still matter, but mostly as a second-wave option for smaller grids, phased industrial loads, coal-site replacement, etc.
- We introduce our weighted scorecard across licensing, financing, fuel, siting, supply chain and engagement (Exhibit 4- Exhibit 7).
- We like SMR projects with high licensing and financing scores, but those projects are mostly government-backed nuclear or US defense and DOE supported demo (Exhibit 8).
- Fuel supply is a gating item, not a footnote. Many advanced SMRs require HALEU, TRISO, metallic, or molten-salt fuel systems. We rank these fuels from high to low supply certainty and find that for most SMR designs, fuel availability is a gating constraint, not a secondary consideration (Exhibit 11).
- Lastly, we discuss the unavoidable math problem: If SMRs cost ~2-3x an AP1000 on a per-kW basis yet deliver less than 1/3 the capacity, the economics demand justification (Exhibit 12). SMRs may retain valid use cases: small grid deployment, coal-site replacement, remote power, defense, and industrial heat, but these are far narrower theses than the AI trade that the market is pricing in.

The near-term trade is existing nuclear: uprates, restarts and life extension. That is where the megawatts are real, the assets are known and the customer need is immediate. Constellation Energy (CEG) and Vistra (VST) are the first major beneficiaries and are working on several large restart projects. Positive news about commercial deployment of the BWRX-300 is a clear benefit for GEV, given its ownership of the GEV Hitachi Nuclear Energy joint ventures that are leading BWRX-300 projects such as the first unit now under construction at Ontario Power Generation (OPG)'s Darlington site.

The longer-term trade is uranium because CCJ and KAP do not need SMRs to work to win. SMRs will not be dead. They are just not the answer to every AI power question.

BERNSTEIN TICKER TABLE

Ticker	Rating	1 Jul 2026		Price Target	TTM Rel. Perf.	Reported EPS				Reported P/E (x)		
		Cur	Closing Price			Cur	2025A	2026E	2027E	2025A	2026E	2027E
CCJ (Cameco)	O	USD	97.39	135.00	14.0%	CAD	1.35	1.66	2.81	30.5	36.5	28.2
CCO.CT (Cameco)	O	CAD	144.56	188.00	19.3%	CAD	1.35	1.66	2.81	30.4	36.4	28.1
KAP.LI (Kazatomprom)	O	USD	70.50	81.00	44.0%	KZT	2,200.00	2,501.52	2,523.28	15.3	13.4	13.3
KZAP.KZ	O	KZT	34,095	39,681	32.6%	KZT	2,200.00	2,501.52	2,523.28	15.5	13.6	13.5
GEV (GE Vernova)	O	USD	1,134.35	1,206.00	98.1%	USD	17.70	25.21	26.05	64.1	45.0	43.5
CEG (Constellation Energy)	O	USD	236.50	296.00	(41.4)%	USD	7.40	11.82	12.67	32.0	20.0	18.7
VST (Vistra Corporation)	O	USD	153.16	181.00	(38.8)%	USD	2.18	9.44	11.70	70.4	16.2	13.1
SPX			7,483.24									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

KAP.LI, KZAP.KZ estimate is Adjusted EPS; CCJ, CCO.CT valuation is EV/EBITDA (x); KAP.LI, KZAP.KZ valuation is Adjusted P/E (x);

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

The near-term trade is in existing nuclear, and CEG and VST are positioned as early major beneficiaries, with multiple large restart projects underway. For GEV, the key upside comes from successful commercial deployment of the BWRX-300, given its stake in the GEV - Hitachi Nuclear Energy JVs leading projects like the first unit at OPG's Darlington site.

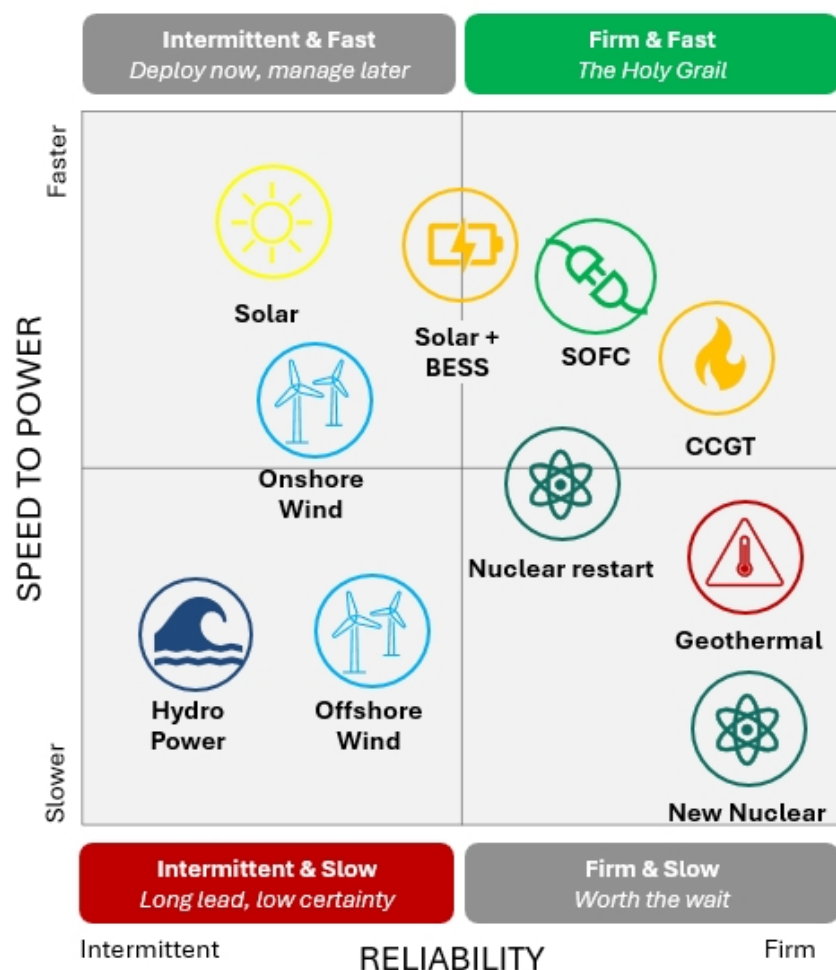
AP300 (and to a lesser degree e-Vinci) is another leading candidate for successful commercial SMR deployment by Westinghouse and CCJ owns 49% of Westinghouse. We ascribe no value to a success-case SMR in our valuation, and thus it is additional upside.

DETAILS

We remind investors that SMRs (a form of new nuclear) are base load, near-zero emissions but are challenged by speed to power. We research each energy source in-depth here:

[The American Energy Transition: Grid, Gas, and Green - Sector Initiation - Power and Energy Transition](#)

EXHIBIT 1: **New Nuclear (including SMR) benefits from reliability but suffers from speed to power**



Source: Bernstein analysis

We also contrast new-build SMRs (the focus of this note) with conventional reactor restarts. Three nuclear projects are confirmed to re-start. In addition, Taiwan could add ~4GW if Kuosheng and Maanshan restart, while Japan has 10+ additional operable reactors that are in the restart process. Reactor re-starts (and the resultant increase in uranium demand) is a more important driver of uranium price in the coming years than SMRs ultimately.

EXHIBIT 2: **We manually add planned restarts to existing nuclear capacity that have been announced**

Plant	Operator	Capacity (MW)	State	In-service
Palisades	Holtec	800	MI	4Q25
Crane/ 3 Mile Island	Constellation	835	PA	1Q28
Duane Arnold	NextEra	615	IA	4Q28
2030 Total		2,250		

Source: Company and news reports, Bernstein analysis and estimates

THE SMR LANDSCAPE

Some definitions. We follow convention and argue that SMRs (Small Modular Reactors) are fission-based nuclear power plants intended to generate electricity at a scale of <300 MW (specifically MWe). Recall that typical reactors are 1,000 MW (1 GW) such as the Westinghouse AP1000. Also recall that MWe is megawatts of electricity equivalent - a measure of power (the ability to generate energy, in this case electrical energy). We also quote MWth or megawatts of thermal equivalent (the ability to generate thermal energy). A good rule of thumb is a reactor's MWe is roughly a third of its MWth.

A large number of SMR projects exist across the world in different stages of maturity (Exhibit 3). We would expect the vast majority to either not move forward or to move forward in the mid 2030s at best.

We highlight that SMRs can be non-commercial (tests, pilots, demonstrations) or commercial (generating electricity from which revenue is earned). As equity investors, our focus is on the latter.

We also remind investors that non-commercial SMR technology has been deployed at scale by various militaries (see the appendix on nuclear vessels). There are currently two civilian operable projects: Rosatom's KLT-40s (providing power to a remote community by leveraging nuclear icebreaker technology) and CNNC's HTR-PM (a demonstration plant initially that has been connected to the grid).

EXHIBIT 3: **The Atlas of SMRs**

Source: WNA, Company reports, Bernstein analysis

The NEA (Nuclear Energy Agency) was founded by the predecessor of the OECD and is an intergovernmental agency with 34 countries holding membership.

The NEA hosts an SMR dashboard that follows all but a few of the global SMR projects we are aware of. Currently the dashboard hosts details for 79 projects (Exhibit 4) totaling 30+ GWth of projects. If we use a typical rule of thumb of 3:1 GWth to GWe and imagine that SMR nuclear capex is ~\$10 bln/GW, the implication is \$100 bln of project capex to deliver the equivalent of 10 typical-scale power plants. We remind investors that commercial SMR doesn't currently exist (we discuss military SMRs in an appendix).

We highlight two projects which are not yet included in NEA: NuScale's RoPower and multiple BWRX-300 fleet projects beyond Darlington.

EXHIBIT 4: The 79 SMRs that NEA tracks

Name	Design Organization	Headquarter	Thermal Power (MWe)	Concept
4S	Toshiba Energy Systems & Solutions Corporation	Japan	30	Metal-cooled
ACP100	China National Nuclear Corporation	China	385	Water-cooled
ACP50S	China General Nuclear Power Group	China	200	Water-cooled
A-HTR-100	Eskom	South Africa	100	Gas-cooled
Aliseas HTGR	Aliseas Engineering	Netherlands	70	Gas-cooled
AP300 SMR	Westinghouse Electric Company	United States	990	Water-cooled
ARC-100	ARC Clean Technology	United States	286	Metal-cooled
Aurora Powerhouse	Oklo	United States	205	Metal-cooled
BANR	BNX Technologies	United States	150	Gas-cooled
Blossom Energy SMR	Blossom Energy	Japan	180	Gas-cooled
Blue Capsule	Blue Capsule Technology	France	150	Metal-cooled
BREST-OD-300	N.A. Dollezhal Research and Design Institute of Power Engineering	Russia	700	Metal-cooled
BWRX-300	GE Vernova Hitachi Nuclear Energy	United States	870	Water-cooled
Cai-30	Gorgé-Calogina	France	30	Water-cooled
CAHEM	Argentina's National Atomic Energy Commission	Argentina	100	Water-cooled
CAWB	Copenhagen Atomics	Denmark	100	Molten salt-cooled
CFRSMR	Saltfloss Energy	Denmark	250	Molten salt-cooled
DF300	Dual Fluid Energy	Canada	600	Metal-cooled
DFBR-1	Deep Fusion	United States	50	Water-cooled
DF-CDE	Dual Fluid Energy	Canada	0	Metal-cooled
EN2	General Atomics	United States	500	Gas-cooled
Energy Well	Research Centre Řež	Czechia	20	Molten salt-cooled
ENTRA1 Energy Plants	NuScale Power	United States	250	Water-cooled
eVinci	Westinghouse Electric Company	United States	15	Heat pipe-cooled
FLEX	MoltenFLEX	United Kingdom	60	Molten salt-cooled
FMR	General Atomics	United States	100	Gas-cooled
GTHTR300	Japan Atomic Energy Agency	Japan	600	Gas-cooled
Halo	Hadron Energy	United States	35	Water-cooled
HAPPY200	China's State Power Investment Corporation (SPIC)	China	200	Water-cooled
HECTAR	Korea Atomic Energy Research Institute	Korea	90	Gas-cooled
Hermes	Kairos Power	United States	35	Molten salt-cooled
Hermes-2	Kairos Power	United States	125	Molten salt-cooled
HEXANA	HEXANA	France	800	Metal-cooled
HTGR-POLA	National Centre for Nuclear Research	Poland	30	Gas-cooled
HTRM-100	Stratek Global	South Africa	100	Gas-cooled
HTR50S	Japan Atomic Energy Agency	Japan	50	Gas-cooled
HTR-PM	Tsinghua University Institute of Nuclear and new Energy Technology	China	500	Gas-cooled
HTR	Japan Atomic Energy Agency	Japan	30	Gas-cooled
IMSR	Terrestrial Energy	United States	884	Molten salt-cooled
I-SMR	Development Agency Korea Hydro & Nuclear Power	Korea	520	Water-cooled

Name	Design Organization	Headquarter	Thermal Power (MWe)	Concept
Jimmy	HTR	France	20	Gas-cooled
Kaleidos	Radiant Industries	United States	4	Gas-cooled
KLT-40S	OKBM Afrikantov	Russia	300	Water-cooled
KRONOS MMR	NANO Nuclear Energy	United States	10	Gas-cooled
LDR-50	Steady Energy	Finland	50	Water-cooled
LEUNR	Canadian Space Mining Corporation	Canada	0	Water-cooled
LFR-AS-200	newcleo	France	480	Metal-cooled
LFR-AS-30	newcleo	France	90	Metal-cooled
LFR	Flibe Energy	United States	600	Molten salt-cooled
LOKI MMR	NANO Nuclear Energy	United States	5	Gas-cooled
MM-1	MobileNuclear Energy	United States	1	Gas-cooled
MoveUx	Toshiba Energy Systems & Solutions Corporation	Japan	10	Heat pipe-cooled
MSR-1	Natura Resources and Abilene Christian University	United States	1	Molten salt-cooled
MSR-100	Natura Resources	United States	250	Molten salt-cooled
Natrium Reactor plant	TerraPower	United States	420	Metal-cooled
NHR200-II	Tsinghua University Institute of Nuclear and new Energy Technology	China	200	Water-cooled
Nuclearis N1	Nuclearis Energy	United States	42	Water-cooled
NUWARD SMR	NUWARD	France	1000	Water-cooled
ODIN	NANO Nuclear Energy	United States	5	Molten salt-cooled
Otrera 2x300	Otrera New Energy	France	600	Metal-cooled
Project Pele	BWA Technologies	United States	0	Gas-cooled
PWR-20	Last Energy	United States	80	Water-cooled
R1	Antares Nuclear	United States	1	Heat pipe-cooled
RITM-200M	OKBM Afrikantov	Russia	396	Water-cooled
RITM-200N	OKBM Afrikantov	Russia	190	Water-cooled
RITM-200S	OKBM Afrikantov	Russia	396	Water-cooled
RR SMR	Rollis-Royce SMR	United Kingdom	1358	Water-cooled
SALUS-100	Korea Atomic Energy Research Institute	Korea	267	Metal-cooled
SEALER-One	Blykalla	Sweden	70	Metal-cooled
SMART100	Korea Atomic Energy Research Institute	Korea	365	Water-cooled
SMR-300	Holtec International	United States	1050	Water-cooled
SSR-W	Moltex Energy	Canada	750	Molten salt-cooled
START	Transmutex	Switzerland	600	Metal-cooled
Stellarium	Stellarium Design	France	1080	Molten salt-cooled
TEPLATOR	University of West Bohemia and the Czech Technical University	Czechia	50	Water-cooled
ThorCon 500	ThorCon International	Singapore	1114	Molten salt-cooled
Thorizon One	Thorizon	Netherlands	250	Molten salt-cooled
XAMR	NAAREA	France	80	Molten salt-cooled
Xe-100	X-energy	United States	200	Gas-cooled

Source: NEA; Bernstein analysis

The majority of the projects on the dashboard are unlikely to proceed before the mid 2030s and so we desire to screen the projects to focus on the quicker or more likely ones. We start with the NEA dashboard because it gives us a common language across very different reactor designs, countries, regulators and business models. We convert that framework into a 100-point score. We give higher weighting to licensing and financing (25% each) as they are the gates where projects move from concept to execution.

EXHIBIT 5: We identify six key factors and assign differentiated weightings to each

Category	Weight %	Why it matters
Licensing	25	Hard gate: cannot build/operate without authorization
Financing	25	FOAK projects often die without credible capital
Fuel	15	High-weight bottleneck, especially for HALEU/TRISO/metallic/molten-salt fuels
Siting	15	No real site = no real project
Supply Chain	15	Nuclear-grade vessels, forgings, fabrication, QA and EPC remain bottlenecks
Engagement	5	Helpful, but does not substitute for licence, fuel, site, money or construction

Source: NEA, Bernstein research and analysis

EXHIBIT 6: **Top SMR projects globally**

Rank	Design	Design Organisation	Status	Use case	Type	HQ	Fuel	Mwe
1	HTTR	Japan Atomic Energy Agency	Operable	Research	HTGR (High temperature Gas-cooled)	Japan	LEU / HALEU range	0 (research only)
2	HTR-PM	Tsinghua University Institute of Nuclear and New Energy Technology	Operable	Demo/Commercial	HTGR	China	HALEU	210
3	KLT-40S	OKBM Afrikantov	Operable	Demo	PWR	Russia	HALEU-range LEU	35 each, 70 total
4	CAREM	Argentina's National Atomic Energy Commission	Under Construction	Demo	PWR	Argentina	LEU	30
5	ACP100	China National Nuclear Corporation (CNNC's Linglong-1)	Operable	Demo/Commercial	PWR	China	LEU	125
6	Hermes	Kairos Power	Under Construction	Demo	FHR (Fluoride salt-cooled High-temp Reactor)	US	HALEU	14
7	Project Pele	BWX Technologies	FID	Demo	HTGR	US	HALEU	1.5
8	RITM-200N	OKBM Afrikantov	FID	Commercial	PWR	Russia	HALEU-range LEU	55
9	BREST-OD-300	N.A. Dollezhal Research and Design Institute of Power Engineering	Under Construction	Demo	LFR (Lead-cooled fast reactor)	Russia	Non-standard fast-reactor fuel	300
10	BWRX-300	GE Vernova Hitachi	Under Construction	Commercial	BWR	US	LEU	300
11	RITM-200S	OKBM Afrikantov	FID	Commercial	PWR	Russia	HALEU-range LEU	58
12	Xe-100	X-energy	Pre-investment	Commercial	HTGR	US	HALEU	80 each; 320 total
13	Hermes 2	Kairos Power	Under Construction	Demo	FHR (Fluoride salt-cooled High-temp Reactor)	US	HALEU	50
14	SMR-300	Holtec International	Pre-investment	Commercial	PWR	US	LEU	320
15	RR SMR	Rolls-Royce SMR	Pre-FID	Commercial	PWR	UK	LEU	470
16	MSR-1	Natura Resources and Abilene Christian University	Under Construction	Research	MSR	US	Non-standard molten-salt fuel	0 (research only)
17	Ropower	Nuscale	FID	Commercial	PWR	US	LEU	77 each; 462 total
18	Sodium	TerraPower	Under Construction	Demo	SFR (Sodium-cooled fast reactor)	US	HALEU	345-500
19	ENTRA1 Energy Plants	Nuscale	Pre-investment	Commercial	PWR	US	LEU	345-500
20	ACPR50S	China General Nuclear (CGN)	Pause	Demo	PWR	China	LEU	60

Source: WNA, NEA, Bernstein research and analysis

SCORING THE SMR PROJECTS TOWARDS A LEADERBOARD

A reactor license without committed financing is ultimately inert - construction permits and NRC approvals mean little if project sponsors cannot close on capital structures and secure offtake agreements. The cancellation of NuScale's CFPP project is one example: despite years of NRC licensing progress and DOE cost-sharing support, the project was cancelled in 2023 when anchor customer UAMPS could not secure enough subscribers at viable power prices and PPAs.

Fuel, siting and supply chain matter next (15% each) because they determine whether a licensed project can actually be built, loaded and repeated.

EXHIBIT 7: Our SMR scoring for the top 20 SMR projects

Rank	Design	Design Organisation	Status	Use case	Type	HQ	Licensing	Siting	Financing	Supply Chain	Engagement	Fuel Criteria	Overall Score
1	HTTR	Japan Atomic Energy Agency	Operable	Research	HTGR (High temperature Gas-cooled)	Japan	C6	C6	C5	C5	C6	C6	94.0
2	HTR-PM	Tsinghua University Institute of Nuclear and New Energy Technology	Operable	Demo/Commercial	HTGR	China	C6	C6	C6	C5	C2	C6	93.5
3	KLT-40S	OKBM Afrikantov	Operable	Demo	PWR	Russia	C6	C6	C5	C5	C3	C6	90.8
4	CAREM	Argentina's National Atomic Energy Commission	Under Construction	Demo	PWR	Argentina	C5	C6	C5	C5	C6	C5	88.0
5	ACP100	China National Nuclear Corporation (CNNC's Linglong-1)	Operable	Demo/Commercial	PWR	China	C5	C6	C5	C5	C6	C4	84.3
6	Hermes	Kairos Power	Under Construction	Demo	FHR (Fluoride salt-cooled High-temp Reactor)	US	C5	C6	C5	C5	C6	C3	80.5
7	Project Pele	BWX Technologies	FID	Demo	HTGR	US	C5	C5	C5	C4	C4	C5	80.0
8	RITM-200N	OKBM Afrikantov	FID	Commercial	PWR	Russia	C5	C5	C5	C4	C4	C4	76.3
9	BREST-OD-300	N.A. Dollezhal Research and Design Institute of Power Engineering	Under Construction	Demo	LFR (Lead-cooled fast reactor)	Russia	C5	C6	C5	C5	C1	C3	75.5
10	BWRX-300	GE Vernova Hitachi	Under Construction	Commercial	BWR	US	C3	C4	C6	C4	C6	C5	69.5
11	RITM-200S	OKBM Afrikantov	FID	Commercial	PWR	Russia	C3	C4	C6	C5	C2	C4	65.3
12	Xe-100	X-energy	Pre-investment	Commercial	HTGR	US	C3	C4	C6	C4	C6	C3	62.0
13	Hermes 2	Kairos Power	Under Construction	Demo	FHR (Fluoride salt-cooled High-temp Reactor)	US	C5	C5	C2	C4	C6	C3	57.0
14	SMR-300	Holtec International	Pre-investment	Commercial	PWR	US	C3	C4	C4	C4	C6	C4	55.8
15	RR SMR	Rolls-Royce SMR	Pre-FID	Commercial	PWR	UK	C3	C4	C4	C4	C6	C4	55.8
16	MSR-1	Natura Resources and Abilene Christian University	Under Construction	Research	MSR	US	C5	C5	C3	C3	C5	C2	54.5
17	Ropower	Nuscale	FID	Commercial	PWR	US	C3	C4	C3	C4	C5	C5	52.5
18	Sodium	TerraPower	Under Construction	Demo	SFR (Sodium-cooled fast reactor)	US	C3	C4	C4	C4	C6	C3	52.0
19	ENTRA1 Energy Plants	Nuscale	Pre-investment	Commercial	PWR	US	C3	C4	C4	C4	C6	C3	52.0
20	ACPR50S	China General Nuclear (CGN)	Pause	Demo	PWR	China	C3	C1	C5	C4	C1	C4	48.0

Source: NEA, IAEA, WNA, Bernstein research and analysis

The top three designs, along with ACP-100 (ranked 5th), prove that small and advanced reactors can be licensed and operated, but they are also bespoke, state-supported, slow, and hard to translate into a repeatable Western commercial case study.

None is, or will be, replicated as a Western First-of-a-Kind (FOAK) project financed on commercial terms to serve data-center load.

HTR-PM is a China state-backed Gen-IV demonstration; HTTR is a Japanese research reactor, not a commercial power plant. KLT-40s is a Russian floating nuclear plant for a remote Arctic grid.

The timing is another problem. HTR-PM (China) took ~11 years to commercial operation. KLT-40s took ~13 years from construction start to commercial operation. HTTR took about 7.5 years from construction to first criticality and more than a decade to operation. Cost transparency is limited, but they would cost around \$500 mln to \$1 bln.

Not included on our list is the RoPower project (Romania) in which an existing coal-fired power plant is to be replaced with 6 SMRs using Nuscale technology. The project reached FID in 2026 but arguably the FID is a conditional FID: "conditions whose fulfillment is the basis for the feasibility of the project and which, consequently, represent mandatory requirements that condition the feasibility of the project" ([Romania's Coal-to-NuScale SMR Conversion Secures FID, Moves Into Implementation with Caveats](#)). Romania's Prime Minister cites costs of \$6-7 bln ([Final Investment Decision taken for Romanian SMR project - World Nuclear News](#)), or \$13-15 bln/GWe (roughly 3x the cost of Chinese-installed CAP1000 and roughly the cost of other SMR projects as well as the cost-plagued US-installed Vogtle AP1000).

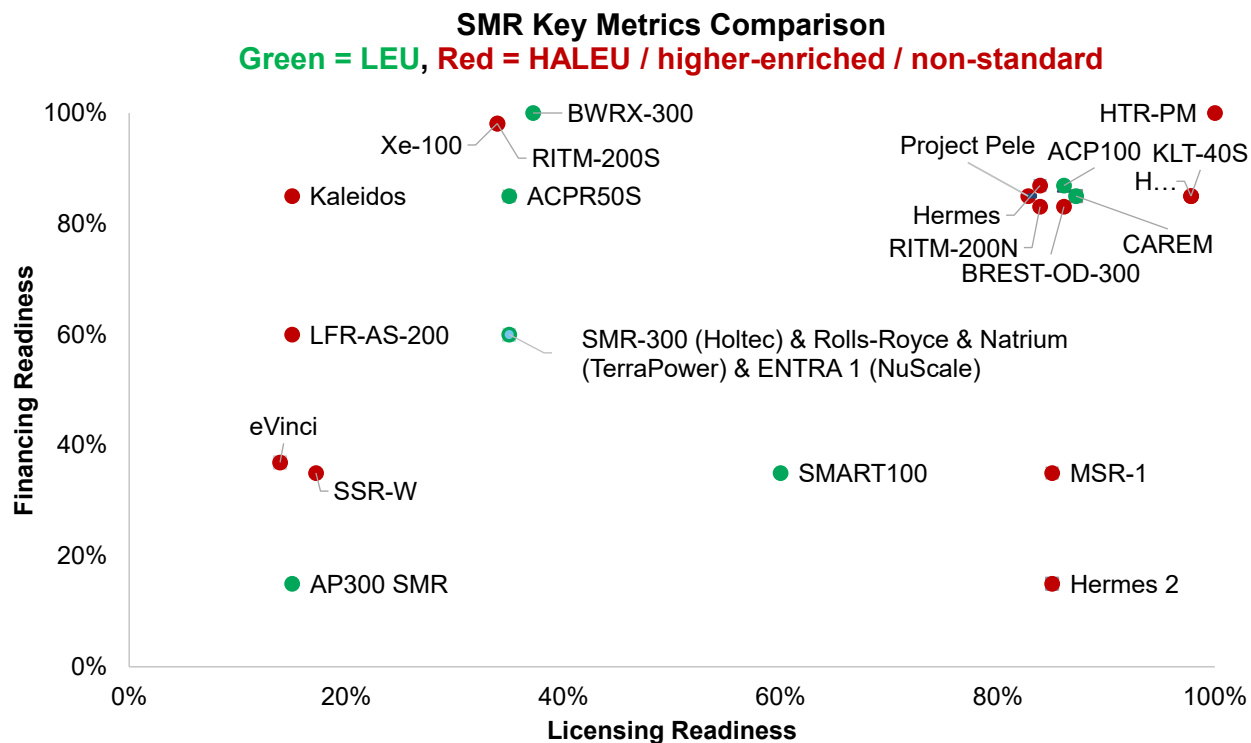
GEV-Hitachi's BWRX-300

UzAtom (Uzbekistan) has poured concrete anticipating the installation of the RITM-200N (a terrestrial SMR modified from the RITM-2000 already deployed on Russian icebreakers). An initial plan of six SMRs was modified to a plan of two 1-GW nuclear power plants and a pair of RITM-200Ns.

A COMBINATION OF FINANCING AND LICENSING IS A SWEET SPOT

The upper-right SMR projects are backed by China, Russia, Japan, Argentina, US Defense, and DOE.

EXHIBIT 8: **We like the upper-right quadrant but they are mostly state-backed SMR projects.**



Source: NEA, IEA, Bernstein research and analysis

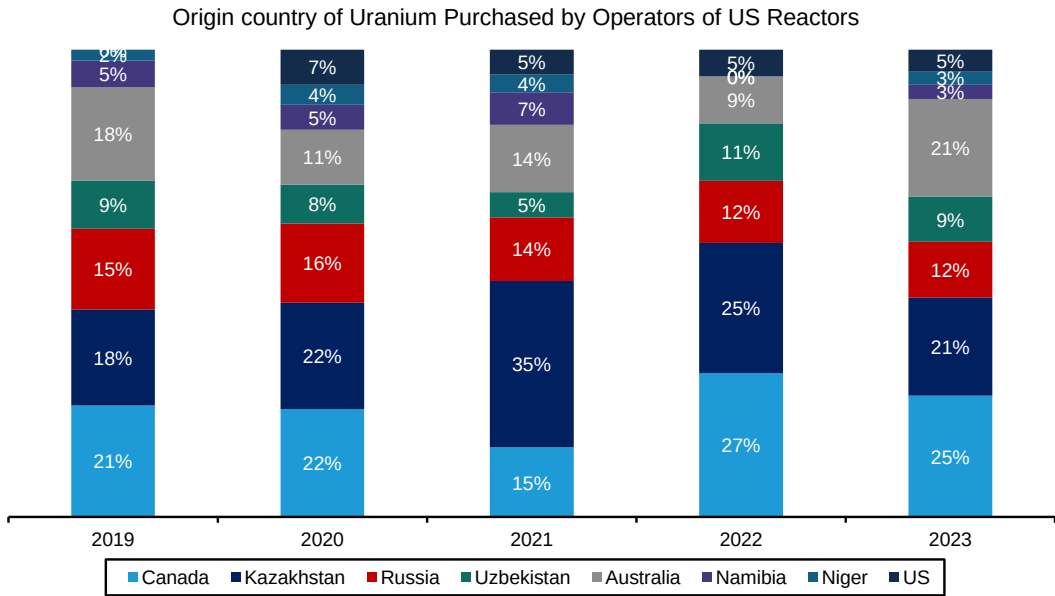
SMRS RELY ON MORE ENRICHED FUEL AND OCCASIONALLY NEWER FORMULATIONS

Those fuels sound like sci-fi, but aren't: Some SMR projects like X-energy's Xe-100 require TRISO fuel, which not only uses scarce HALEU but also wraps each HALEU kernel in multiple concentric layers of pyrolytic carbon and silicon carbide - a mini containment vessel inside every particle. **But on a more unconventional fuel pathway is molten salt reactor (MSR) fuel:** Natura Resources and Abilene Christian University's MSR program uses HALEU dissolved directly into a circulating molten salt mixture. so the core is literally a moving radioactive liquid, with the fuel, coolant and chemistry all happening in one flowing salt loop.

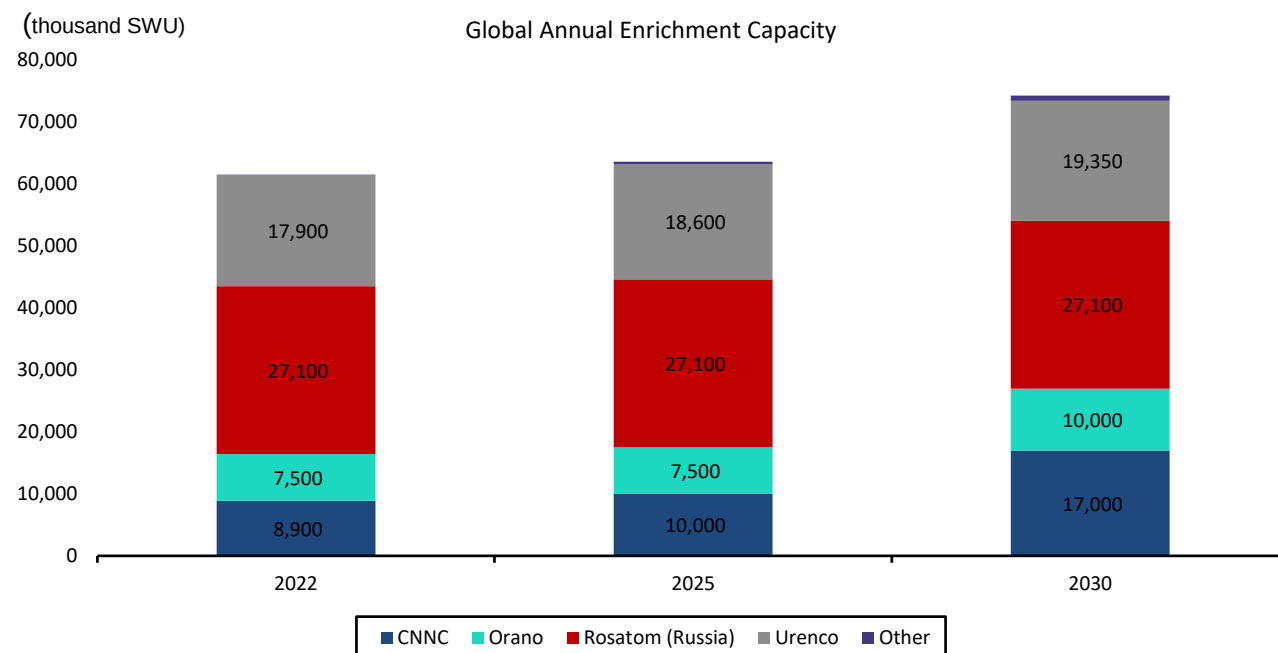
The DOE is seeking to make 21 metric tons (MT) of HALEU available by Jun 2026 (3MT by Sep 2024, 8MT by Dec 2025, 10MT by Jun 2026). The DOE named five initial HALEU recipients: TRISO-X, Kairos, Radiant, Westinghouse and Terrapower. But DOE allocation doesn't guarantee commercial success.

In May 2024, the US passed a law banning the import of Russian-produced low-enriched uranium (LEU), with the ban fully taking effect in August, though waivers are available until 2028. Russia, on November 15, 2024, retaliated by imposing temporary restrictions on exporting enriched uranium to the US. This is a direct counter - measure to the US sanctions.

EXHIBIT 9: **US purchased 40.5M lbs of U3O8 in 2022, and 12% is from Russia with very low price (\$31/lb in 2023 vs. \$51/lb from Australia, \$43/lb from Canada and Kazakhstan)**



Source: EIA, Company reports, Bernstein analysis

EXHIBIT 10: **Russia still accounts for ~40% of global enrichment capacity in 2030**

Source: WNA, company reports, Bernstein analysis

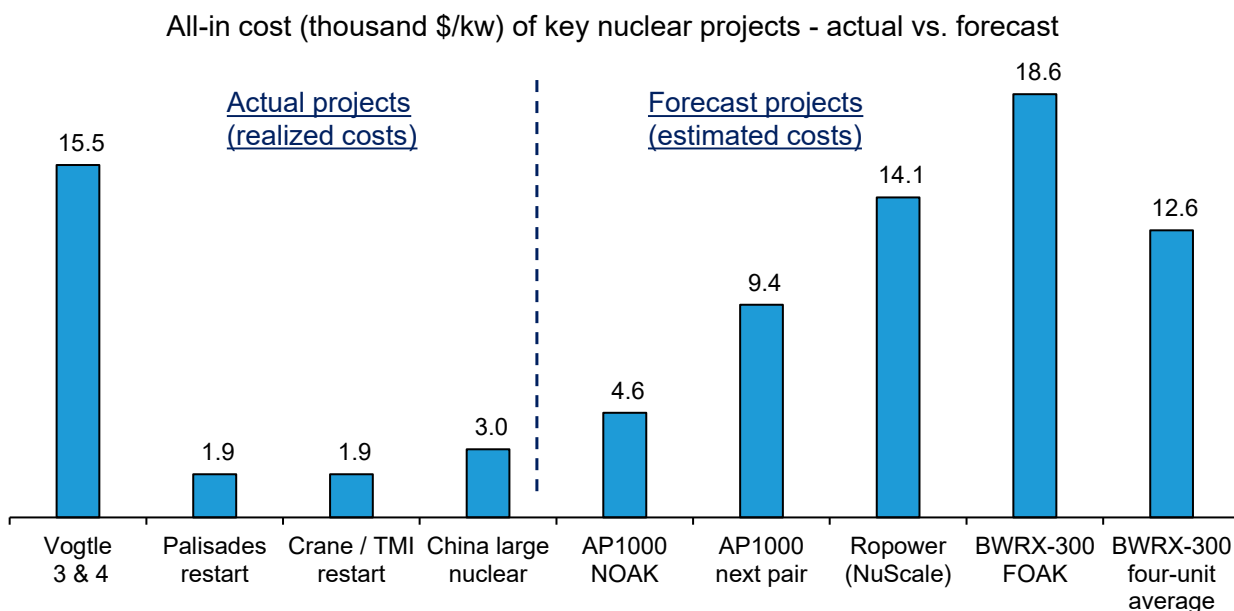
EXHIBIT 11: **Besides those that require LEU, most SMRs projects have fuel risk.**

Fuel pathway	Risk level	Designs	Why
Conventional LEU / LWR fuel	Lowest	BWRX-300, AP300, NuScale, Holtec SMR-300, Rolls-Royce SMR	Existing and proven LWR fuel ecosystem; Risk is more about conversion/enrichment capacity.
HALEU + TRISO	Medium high	X-energy, Kairos, Project Pele, eVinci, Oklo-type pathways	TRISO is technically proven, but commercial scale HALEU supply is still limited
HALEU / metallic / fast-reactor fuel	High	TerraPower Sodium, ARC-type, BREST-type	Needs HALEU+reactor specific metallic fuel qualification and more licenses.
Molten-salt / salt fuel or salt coolant systems	Highest fuel & system risk	MSR-1, SSR-W, some Kairos-type salt-cooled concepts	Risk of fuel chemistry, licensing risk, least standardized commercial pathway.

Source: Company reports, NEA, IEA, Bernstein research and analysis

COST MATTERS, OPPORTUNITY ALSO MATTERS

The purpose of SMRs can be two-fold: (1) a bespoke project suitable for a unique application (i.e., remote small community lacking other forms of electricity) or (2) a game-changing NOAK (Nth of a Kind) technology where the initial investments (i.e., high cost projects) pave the way to a competitive source of electricity. For the second case to hold, we would argue (1) a commitment to a significant number of projects (10+?), and (2) a modular repeatable proven design. We would argue that a winner-take-all market where just a few SMR designs move forward is most likely to support the second case. Thus we'd "hope" that 79 concepts screened down to 25 projects ultimately leads to perhaps 1-3 winners. We'd argue the BWRX-300 (GE Vernova) and AP-300 (Westinghouse) could be amongst those.

EXHIBIT 12: **Early SMR projects more expensive than re-starts and conventional projects**

Source: WNA, NEA, IAEA, Company reports, Bernstein research and analysis

SOME CAUTIONARY SMR SCHEDULE SLIPS

Lastly, we list a few key SMR projects that have slipped and highlight that as a risk for all projects:

- Perfect score, frozen site - CAREM:** Back to our x-y scatter chart (Exhibit 8), CAREM stands out with both high financing and licensing scores, and using LEU, yet it stands as the cautionary tale. CAREM broke ground in Argentina in February 2014 and is one of the most advanced SMR construction projects globally by physical completion (85% physically complete). However, construction has been halted multiple times after federal funding was cut amid an unstable political environment. Argentina's refreshed nuclear policy, unveiled in May 2026, sought to separate political leadership from the country's nuclear operations, but there are no public reports of construction resuming as of Jun 2026.
- Oklo:** Oklo was publicly guiding 2027 as recently as March 2025, and the DOE's Reactor Pilot Program set a criticality by July 4, 2026 milestone that is clearly not being met. The timeline has quietly shifted to late 2027 to early 2028. The NRC combined license application (COLA) has still not been submitted as of mid-2026, which is the critical gating item for commercial operation — yet construction is already underway.
- TerraPower** Natrium's 3-year delay: Construction permitting was only received in March 2026, three years behind the DOE's original 2027 first power target. Nuclear island construction is to begin in 2026 with fuel loading now targeted for 2030 and commercial operations in 2031. The operating license application is not yet submitted, with NRC review of that application still ahead — introducing further schedule risk on what is already a FOAK sodium fast reactor.
- Nuscale's** RoPower - scope reduction & financing risk: FID slipped from Q2 2025 to February 2026, and the approved structure reduced scope to a single 77 MW test module before committing to the full six-unit 462 MW build. As discussed above, the FID feels more conditional than a true FID. First operation is now targeted for 2033, with financing and EPC contracts still to be finalized — prolonging uncertainty around PPA structures and project economics.

SMR FUEL DEMAND CAN DRIVE URANIUM DEMAND UPWARDS

We highlighted the nuclear fuel cycle in [Americas Energy & Transition: Top pushbacks to our Cameco initiation](#). Below is the recap of the enrichment step and how that implies to uranium.

Massive adoption of SMR will create more uranium demand. And here we re-introduce our favorite orange juice analogy.

We start with an analogy of making orange juice from oranges. If I want a glass of orange juice, there is a trade off how to fill it. I can gently squeeze a large number of oranges, or I can greedily and forcefully squeeze a smaller number of oranges. I can trade off the material into the process against the work or effort of squeezing.

The same thing happens in enrichment, causing a potential opportunity for Cameco, which we discuss.

Natural uranium consists of ~0.7% of the U-235 isotope, while the remaining 99.3% is the U-238 isotope.

Isotope separation is a physical technique used to increase the concentration, or 'enrich' one isotope in relation to others. The enrichment plants primarily are centrifuge based (other techniques exist such as diffusion). Because U-235 is lighter than U-238, when the centrifuges are spun rapidly (50,000 - 70,000 rpm) during the enrichment process, it helps separate the heavier U-238 from the lighter U-235. The more centrifuges the gas passes through, the higher the U-235 content it has at the end and thus the higher assay.

Clearly it takes "spinning" or "work" or "effort" or "available equipment" to enrich.

The majority of reactors, which are light water reactors (either Pressurized Water Reactors - PWR or Boiling Water Reactors - BWR), require low-enriched uranium (LEU) as fuel with U-235 increased from the natural 0.7% to 3-5% product assay. 3% product assay corresponds to requirements from older reactors, while 4%-5% is the standard for modern light water reactors.

There is growing interest in enriching uranium to approximately 7% and, for certain special power reactor (Gen IV and SMR), even up to close to 20%, which is classified as high-assay LEU (HALEU). That would take more "spinning".

To describe the work of enrichment, the industry uses the concept of Separative Work Units (SWU) to measure enrichment. We don't like the term SWU but bow to the industry use. SWU is proportional to energy but incorporates also a notion of capacity.

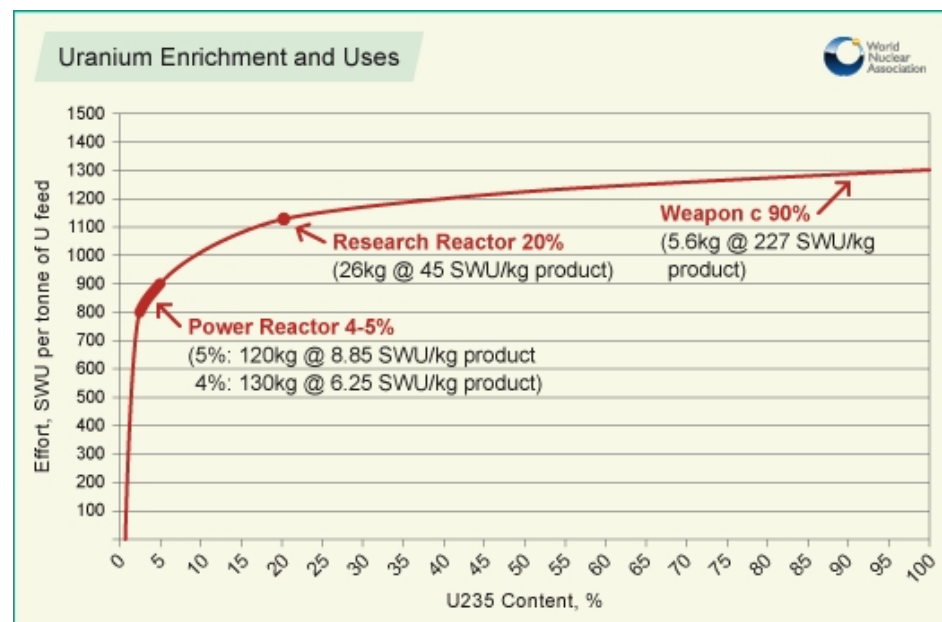
One other concept of SWU is separation. Enrichment takes an input mixture of U-235 and U-238 and separates them into two piles - one with more U-235 and one with less U-235. Clearly if I have two piles and one is richer in U-235, the other (the tailings) must be poorer.

The chart below shows the SWU ("work") to move from natural uranium at 0.7% to various levels of U-235.

Let's say we start with 1,000 kg of input (the yellowcake that has been refined and converted). At 0.7% U-235, it contains 7 kg of U-235 and 993 kg of U-238. If I separate them perfectly, it would take 1,300 SWU (that much work/effort/energy).

If I was satisfied with 5% grade and wanted 1 kg of enriched uranium product produced from yellowcake (at 0.7%), I have a choice. I can be sloppy and discard a lot of my U-235 in the tailings (say 0.5%) [i.e., do a bad job squeezing the oranges and end up with a pile of poorly squeezed oranges with juice left in them]. That would require 45 kg of yellowcake but only consume 13 SWU of work. If I am diligent and discard 'cleaner' tailings (say only 0.1% U-235), I would need 16 kg of yellowcake (a third less!) but consume 28 SWU of work (twice the work). If I need a certain amount of enriched product for my reactors but I have limited enrichment facilities (limited SWU), then I have to consume more uranium to get there.

EXHIBIT 13: **Enrichment takes work**

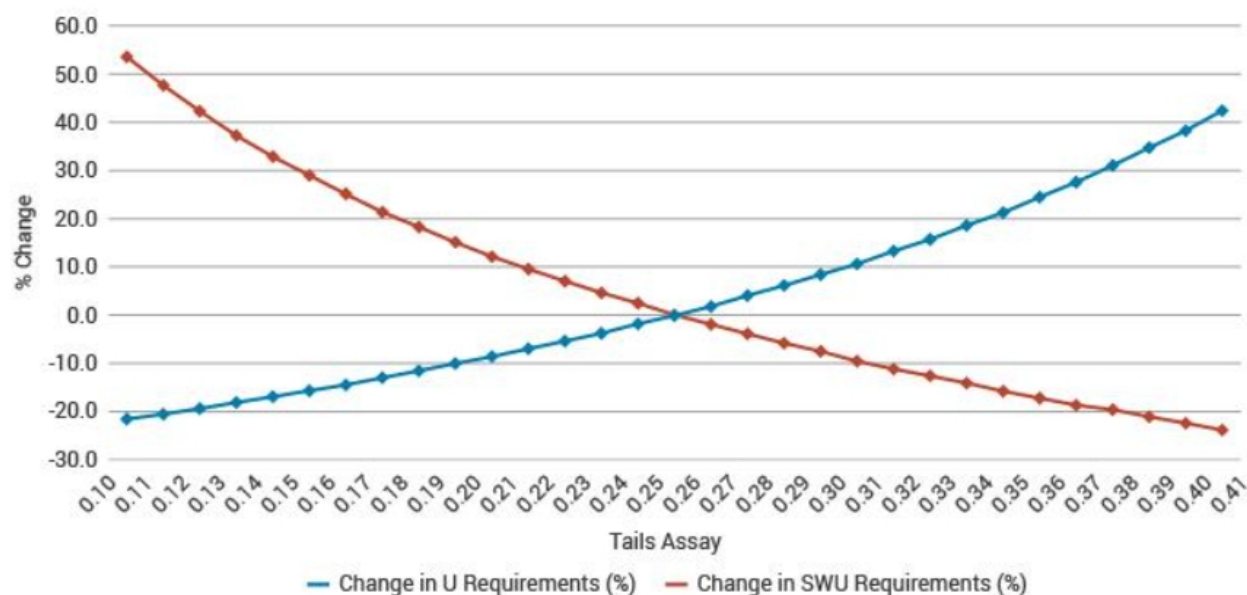


Source: World Nuclear Association

When the market has limited SWU or the enrichment capacity is reached, overfeeding follows. Overfeeding happens when enrichers use more uranium feedstock (UF₆) than usual, while leaving higher amounts of U-235 in the waste, aka producing higher tails assay. More demand for UF₆ leads to increased uranium demand, tightening the supply/demand balance. We can understand this as a trade-off: consuming more natural uranium to ease pressure on SWU supply.

EXHIBIT 14: **We can understand this as a trade-off: consuming more natural uranium to ease pressure on SWU capacity.**

Percentage Variation in Uranium Requirements and Energy Input to Enrichment with Different Tails Assay, from a Base of 0.22% U-235



Source: Company reports, Bernstein analysis

The general formula is **Feed to Product** = $(X_p - X_t) / (X_f - X_t)$, where F is feed, P is product, and T is tail.

At the same time, the world requires higher enriched fuel (higher than the usual 3-5%) for SMR and Gen IV reactors. Higher product and tails assays will together exponentially increase the demand for UF_6 and thus U_3O_8 , shown from the sensitivity table (assuming 0.711% industry - average feed assay).

EXHIBIT 15: **With demand on more enriched feed but with limited SWU available, enrichers will need more UF6 and thus U3O8 to meet such growing demand**

Sensitivity		Tails Assay			
Product Assay	kgU as UF6	0.20%	0.25%	0.30%	0.35%
	3.0%	5.5	6.0	6.6	7.3
	3.5%	6.5	7.1	7.8	8.7
	4.0%	7.4	8.1	9.0	10.1
	4.5%	8.4	9.2	10.2	11.5
	5.0%	8.9	10.3	11.4	12.9
	5.5%	10.0	11.4	12.7	14.3
	6.0%	11.4	12.5	13.9	15.7

kgU as UF6		Tails Assay			
Product Assay	% change	0.20%	0.25%	0.30%	0.35%
	3.0%	-32.6%	-26.7%	-19.2%	-9.7%
	3.5%	-20.6%	-13.3%	-4.3%	7.3%
	4.0%	-8.6%	0.0%	10.7%	24.3%
	4.5%	3.5%	13.3%	25.6%	41.3%
	5.0%	8.8%	26.7%	40.6%	58.4%
	5.5%	23.2%	40.0%	55.5%	75.4%
	6.0%	39.5%	53.3%	70.5%	92.4%

Source: UxC Fuel Cost Calculator, Company reports, Bernstein analysis

We do not explicitly include this factor in our supply/demand balance for yellowcake, but highlight it as a future pinchpoint (which could tighten the markets and benefit CCJ by consuming more UF6 and U3O8).

APPENDIX - WHAT NUCLEAR VESSELS TEACH US ABOUT SMR

The first use of nuclear power was the Trinity nuclear test (July 1945).

The next month (August 6th and 9th 1945), two nuclear bombs would be used against Japan by the United States.

The Soviet Union would follow the US in weapons development and test their first weapon by 1949.

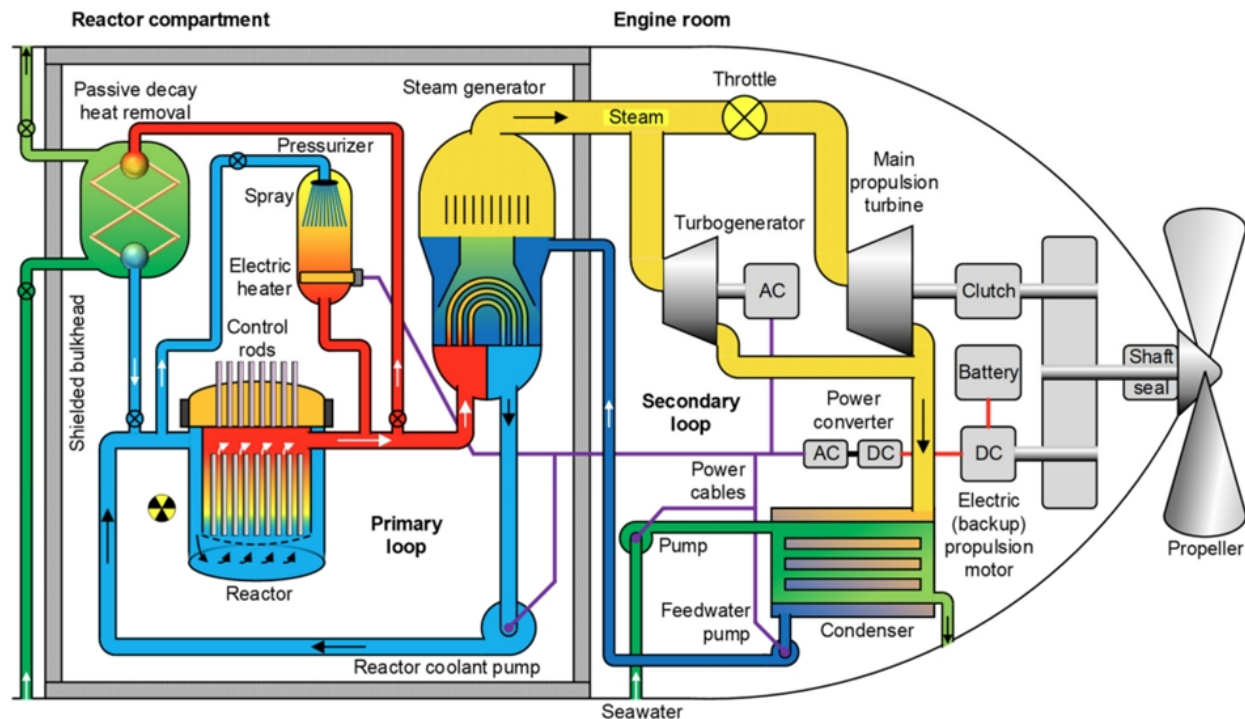
Peace-time uses of nuclear power would follow. By December 1951, EBR-1 (Idaho) would generate sufficient electricity for four light bulbs. 1954 would see the Obninsk power plant generate small scales of industrial electricity.

Calder Hall (United Kingdom) would become the first industrial-scale project in 1956 with 4 reactors and a total of 240 MW of electricity or MWe (roughly a quarter the size of today's typical design such as an AP1000). **The world's first power plant was thus an SMR (<300 Mwe).**

Also in 1951 was the authorization of the USS Nautilus (SSN-571). Authorized in 1951 and delivered in 1955, the Nautilus was the first nuclear-powered boat and submarine.

The Nautilus was powered by an S2W reactor ("S" for submarine, "2" for 2nd generation, and "W" for Westinghouse). It delivered 60-70 MWt (MW of thermal energy).

EXHIBIT 16: The structure of a nuclear submarine SMR system



Source: Holbert, 2025 (CC BY); Bernstein analysis

1958 would see the Soviet's first nuclear submarine. 1961 would see the first nuclear aircraft carriers and cruisers.

On the civilian side, The Soviet Union would launch the first vessel – an icebreaker in 1959 – followed by other merchant vessels from the US (1962), Germany (1969) and Japan (1972).

Civilian ships were challenged not by technology nor economics but rather by litigation, specifically around liability/indemnification around accidents and port entry.

As an aside, earlier nuclear submarines had poorer shielding (with seamen exposed to 173 mrem/year compared to today's seamen at 10 mrem/year). For context, a chest x-ray is 10 mrem while a full-body CT is 1,000 mrem. An hour flying is 1 mrem. A typical American annual exposure budget is 620 mrem (strongly driven by whether your home has radon or thoron). The point being that even **early generation nuclear submarines were safe.**

In terms of reactor design, we show innovation but highlight (a) a small number of 'winners', and (b) a focus on PWR (the first technology that worked was improved upon). **We'd expect the SMR race to have a few winners with the best & simplest technology ultimately being adopted.**

EXHIBIT 17: **The naval vessel SMR evolution**

Purpose	Generation	Manufacturer	Type	Innovation	Vessel	Class	Manufacturer
S	1	W	PWR	Superheated steam	Prototype	Submarine	Westinghouse
S	2	W	PWR		Nautilus	Submarine	Westinghouse
S	1	G	LMR	Liquid sodium cooling	Prototype	Submarine	General Electric
S	2	G	LMR		Seawolf	Submarine	General Electric
S	2	Wa	PWR	Saturated steam	Seawolf	Submarine	Westinghouse
S	3	W	PWR	Horizontal U-tube	Halibut	Submarine	Westinghouse
S	4	W	PWR	New arrangement	Swordfish	Submarine	Westinghouse
S	5	W	PWR	Simplicity	Skipjack	Submarine	Westinghouse
S	3	G	PWR	DFT	Prototype	Submarine	General Electric
S	4	G	PWR		Triton	Submarine	General Electric
S	1	C	PWR		Prototype	Submarine	Combustion Engineering
S	2	C	PWR		Tullibee	Submarine	Combustion Engineering
S	5	G	PWR		Narwhal	Submarine	General Electric
S	6	G	PWR		Los Angeles-class	Submarine	General Electric
S	6	W	PWR		Seawolf-class	Submarine	Westinghouse
S	7	G	PWR		Prototype	Submarine	General Electric
S	8	G	PWR		Ohio-class	Submarine	General Electric
S	9	G	PWR		Virginia-class	Submarine	General Electric
S	1	B	PWR		Columbia-class	Submarine	Bechtel
A	1	W	PWR		Long Beach	Aircraft Carrier	Westinghouse

Source: Company reports; NTI; Bernstein analysis

A sidenote: S2W was initially sodium-cooled (LMR or Liquid Metal Reactor) but moved to PWR (pressurized water reactor). The Soviet Union built perhaps 8 LMR vessels (but settled on PWR). Gas-cooled reactors have been contemplated but not deployed but are more advanced than boiling water reactors or organic moderated reactors.

Another final aspect of nuclear vessels is the level of enrichment. While some nuclear vessels (such as US designs) use HEU (Highly Enriched Uranium) which is weapons-grade, some vessels can use LEU (Low Enriched Uranium) or HALEU. **We'd expect civilian SMRs to rely on LEU and HALEU but not HEU.**

DISCLOSURE APPENDIX

I. REQUIRED DISCLOSURES

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VALUATION METHODOLOGY**Cameco Corp**

We rate Cameco Outperform and value Cameco at \$135/sh (CAD \$188/sh) using a 23x EV/EBITDA multiple on its 2030E EBITDA of CAD\$3.2bln.

NAC Kazatomprom JSC

We rate Kazatomprom Outperform and value Kazatomprom at \$81/sh using a 8x EV/EBITDA multiple on its 2027E EBITDA of KZT 1,368 bln and incorporating additional returns to shareholders.

GE Vernova, Inc

We use a 28x multiple on 2027-28 estimated EBITDA of ~\$10.7B to arrive at our target price of \$1,206/sh.

Constellation Energy Corporation

We use an SOTP valuation running a 14x multiple on 2027-28E Nuclear EBITDA of \$5.7B and a 10x multiple on 2027-28E rest of group EBITDA of \$3.5B to arrive at our target price of \$296/sh.

Vistra Corporation

We use a 8.7x multiple on 2027-28 estimated adjusted EBITDA of \$8.7B to arrive at our target price of \$181/sh.

RISKS**Cameco Corp**

Downside risks to our price target include:

1. Nuclear safety - related risks

Nuclear power safety concerns, often due to reactor mismanagement, can cause delays in permits and construction. This slowdown in the nuclear energy sector adoption directly reduces the demand for Cameco's uranium, hitting its sales and revenue. Public fear and regulatory tightening can further dampen the market.

2. High labor costs

Cameco faces high costs for skilled labor in the nuclear industry. The limited supply of specialized workers drives up wages, squeezing profit margins. Labor - market fluctuations like shortages or strikes can disrupt production and increase expenses.

3. The rise of plutonium & thorium use could reduce demand for uranium

The growing use of plutonium and thorium in Fast Breeder Reactor (FBRs) and Molten Salt Reactor (MSR) threatens Cameco's uranium market share. However, Plutonium's high radioactivity and potential for misuse lead to strict regulations and possible public opposition, and the development of FBR and MSR are still in the early stage with more exposure in Asia. Considering Cameco's small exposure in Asia (17% uranium & 5% of UF₆ sold to Asia in 2024), the impact of this Gen IV reactor development will be limited on Cameco.

4. Inconsistent operation and transparency in Inkai with Kazatomprom

In the Inkai project with Kazatomprom, inconsistent operations can lead to production inefficiencies and unexpected costs. Lack of transparency may cause misunderstandings and disputes between the partners. Both issues can undermine the project's performance and Cameco's returns.

NAC Kazatomprom JSC

Downside risks to our price target include:

1. Nuclear safety - related risks

Nuclear power safety concerns, often due to reactor mismanagement, can cause delays in permits and construction. This slowdown in the nuclear energy sector adoption directly reduces the demand for KAP's major business - uranium, hitting its sales and revenue. Public fear and regulatory tightening can further dampen the market.

2. The rise of plutonium & thorium use could reduce demand for uranium

The growing use of plutonium and thorium in Fast Breeder Reactor (FBRs) and Molten Salt Reactor (MSR) threatens KAP's uranium market share. However, Plutonium's high radioactivity and potential for misuse lead to strict regulations and possible public opposition, and the development of FBR and MSR are still in the early stage with more exposure in Asia.

3. Inconsistent operation and delay of sulfuric acid

Both geopolitical tension and lack of sulfuric acid will impact KAP's production and sales. Geopolitical conflict with China will impact KAP's sales negatively.

GE Vernova, Inc

Key downside risks to our price target include continued negative earnings from offshore wind beyond 2027 from a weak macro, execution risk, and lower overall power demand driving slower backlog growth beyond 2030 in the power business. Their tie to hydrocarbons may present a risk to ESG inclined investors.

Constellation Energy Corporation

Key downside risks to our price target include the inability to contract open capacity at attractive terms after the phase out of tax credits, delays in adding new nuclear generation capacity, and meaningful moderation in PJM capacity prices following the RBP.

Vistra Corporation

Key downside risks to our price target include lower than expected power demand growth from data centers in the ERCOT and PJM regions, high natural gas prices impacting margins, and unfavorable hedging outcomes.

RATINGS DEFINITIONS, BENCHMARKS AND DISTRIBUTION

EQUITY RATINGS DEFINITIONS

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The Bernstein brand rates stocks based on forecasts of relative performance for the next 12 months versus the S&P 500 for stocks listed on the U.S. and Canadian exchanges, versus the Bloomberg Europe Developed Markets Large and Mid Cap Price Return Index EUR (EDME) for stocks listed on the European exchanges and emerging markets exchanges outside of the Asia Pacific region, versus the Bloomberg Japan Large and Mid Cap Price Return Index USD (JPL) for stocks listed on the Japanese exchanges, and versus the Bloomberg Asia ex-Japan Large and Mid Cap Price Return Index (ASIAX) for stocks listed on the Asian

(ex-Japan) exchanges -unless otherwise specified.

The Bernstein brand has three categories of ratings:

- Outperform: Stock will outpace the market index by more than 15 pp
- Market-Perform: Stock will perform in line with the market index to within +/- 15 pp
- Underperform: Stock will trail the performance of the market index by more than 15 pp

Coverage Suspended: Coverage of a company under the Bernstein research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Not Covered (NC) denotes companies that are not under coverage.

Bernstein brand stock ratings are based on a 12-month time horizon.

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The Autonomous brand rates common stocks as indicated below. As our benchmarks we use the Bloomberg Europe 500 Banks And Financial Services Index (BEBANKS) and Bloomberg Europe Dev Mkt Financials Large and Mid Cap Price Ret Index EUR (EDMFI) index for developed European banks and Payments, the Bloomberg Europe 500 Insurance Index (BEINSUR) for European insurers, the S&P 500 and S&P Financials for US banks and Payments coverage, S5LIFE for US Insurance, the S&P Insurance Select Industry (SPSIINS) for US Non-Life Insurers coverage, and the Bloomberg Emerging Markets Financials Large, Mid and Small Cap Price Return Index (EMLSF) for emerging market banks and insurers and Payments. Ratings are stated relative to the sector (not the market).

The Autonomous brand has three categories of common stock ratings:

- Outperform (OP): Stock will outpace the relevant index by more than 10 pp
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- Underperform (UP): Stock will trail the performance of the relevant index by more than 10 pp

Coverage Suspended: Coverage of a company under the Autonomous research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Those denoted as 'Feature' (e.g., Feature Outperform FOP, Feature Under Outperform FUP) are our core ideas.

Not Covered (NC) denotes companies that are not under coverage.

Autonomous brand common stock ratings are based on a 12-month time horizon.

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- Outperform (OP): The total return of the preferred instrument is expected to outperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- Neutral (N): The total return of the preferred instrument is expected to perform in line with preferred securities of other issuers operating in similar sectors or rating categories over the next six months.

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The report may also include reference(s) to published opinions by other Autonomous or Bernstein analysts covering the equity securities of the issuer(s) referenced herein. Please note an investment recommendation for credit instruments published by the author(s) of this report may differ from the published view of the analyst covering equity securities for the issuer(s) contained in this report and vice versa.

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A list of all investment recommendations produced by the author(s) of this report alongside credit ratings history are available upon request.

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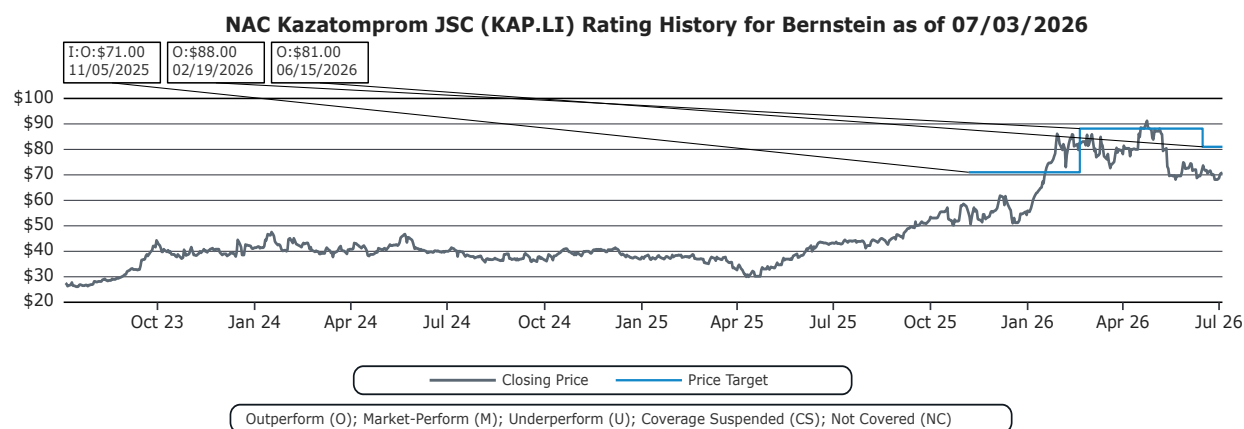
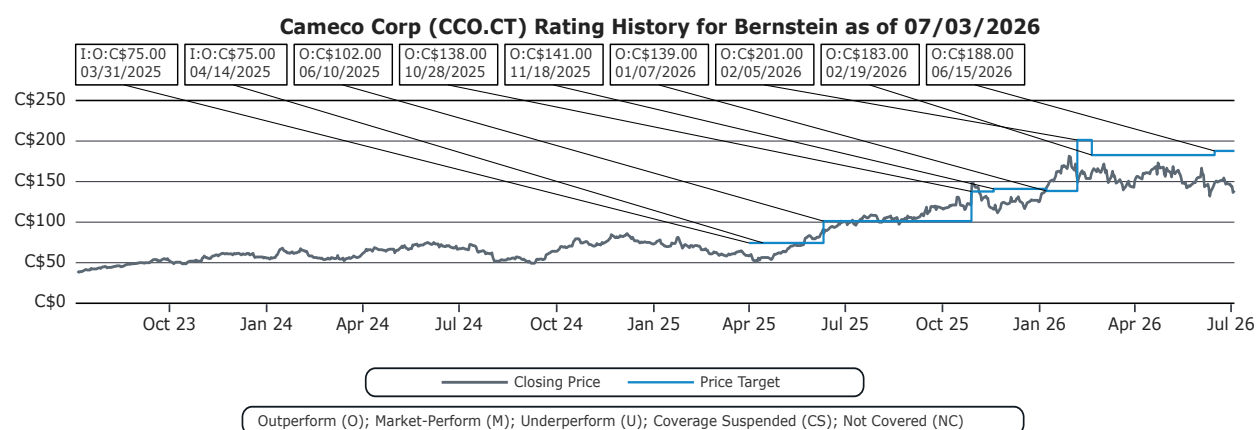
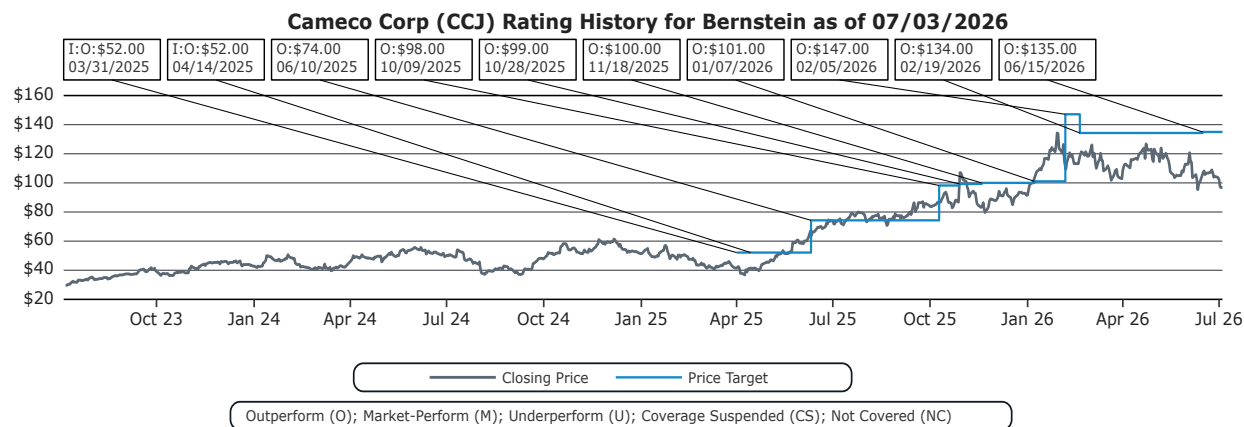
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Equity Rating	Market Abuse Regulation (MAR) and FINRA Rating Category	Global Rating Distribution	Investment Banking Relationships*
Outperform	BUY	51.2%	15.3%
Market-Perform (Bernstein Brand) Neutral (Autonomous Brand)	HOLD	35.8%	16.2%
Underperform	SELL	13.1%	13.6%

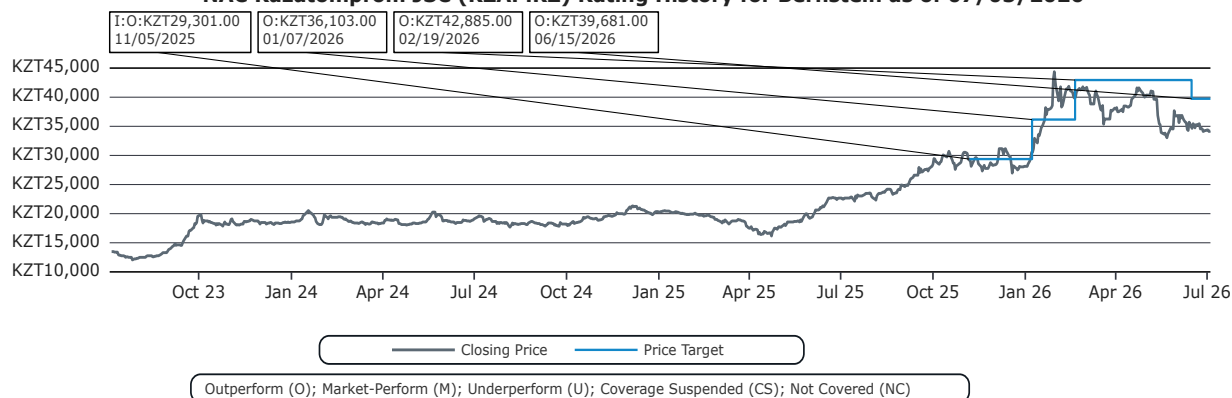
* These figures represent the percentage of companies within each equity rating category for which affiliates of Bernstein have provided investment banking services within the previous 12 months.

As of June 30, 2026. All figures are updated quarterly.

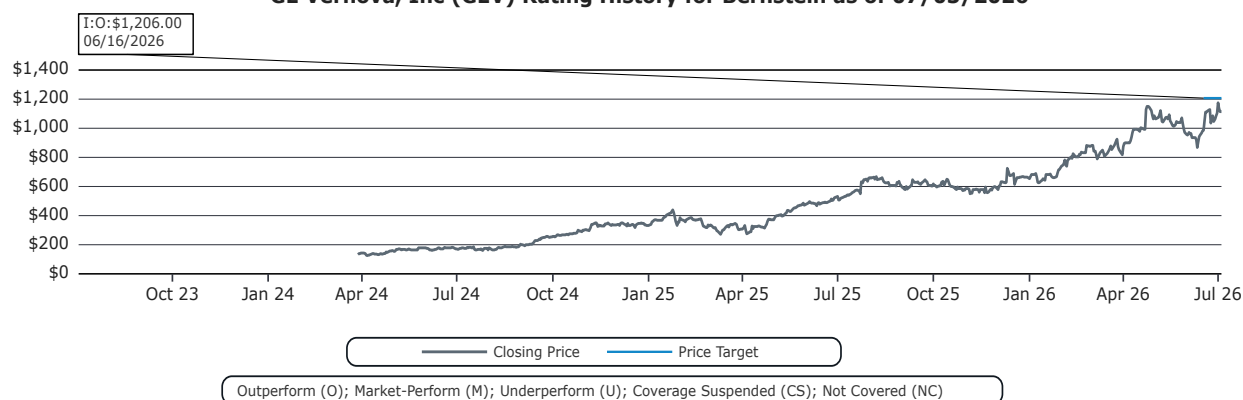
PRICE CHARTS / RATINGS AND PRICE TARGET HISTORY



NAC Kazatomprom JSC (KZAP.KZ) Rating History for Bernstein as of 07/03/2026

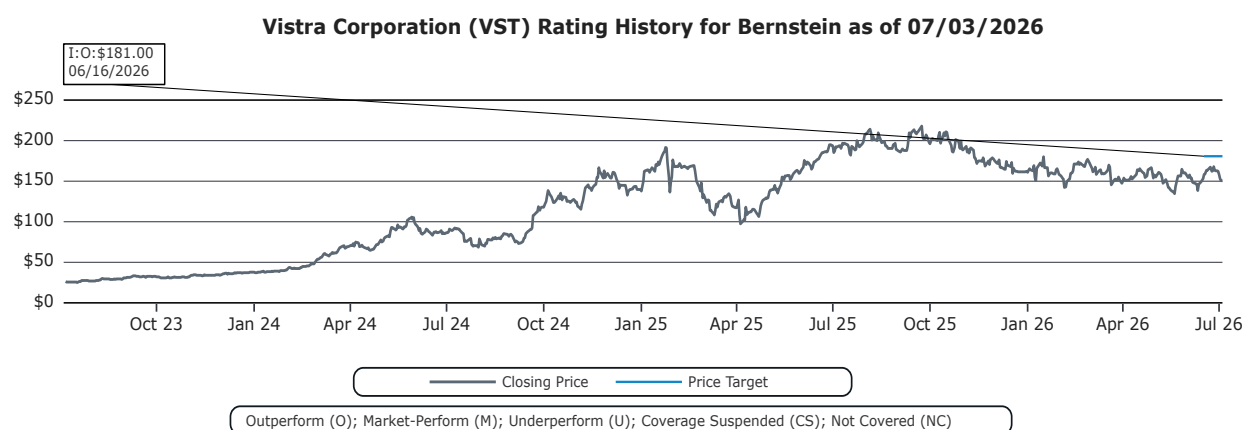


GE Vernova, Inc (GEV) Rating History for Bernstein as of 07/03/2026



Constellation Energy Corporation (CEG) Rating History for Bernstein as of 07/03/2026





All price target and closing price data in the chart(s) above are denominated in the currency noted in the Ticker Table of this report.

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Bernstein and/or affiliates have received compensation for investment banking services in the past twelve months from GE Vernova, Inc.

Bernstein and/or affiliates have received compensation for non-investment banking securities-related products or services in the previous twelve months from the following clients: GE Vernova, Inc and Constellation Energy Corporation.

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